

Fringer: A Finger-Worn Passive Device Enabling Computer Vision Based Force Sensing Using Moiré Fringes

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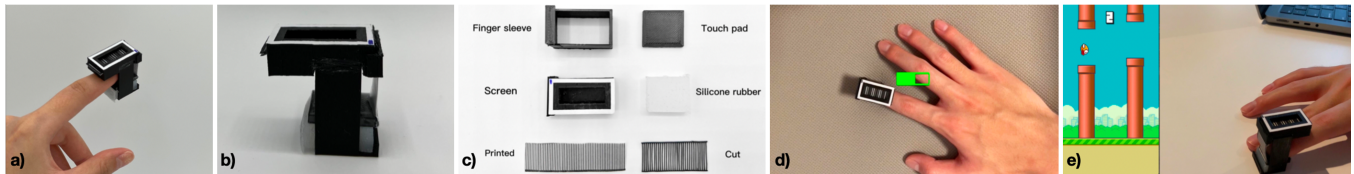


Figure 1: a) top view and b) side view of Fringer. c) Fringer components. Applications: d) a force visualizer for video recording, and e) a force-based game controller

ABSTRACT

Despite the importance of utilizing forces when interacting with objects, sensing force interactions without active force sensors is challenging. We introduce Fringer, a finger sleeve that physically visualizes the force to allow a camera to estimate the force without using any active sensors. The sleeve has stripe-pattern slits, a sliding paper with stripe pattern, and a compliant layer that converts force into sliding paper movements. The patterns of the slit and the paper have different frequencies to create Moiré fringes, which can magnify the small displacement caused by the compliant layer compression for webcams to easily capture such displacement.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

force sensing, force visualization, Moiré patterns, wearable sensor

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1 INTRODUCTION

Haptic interactions, such as touching and pressing, are crucial parts of everyday interactions with objects and devices. Therefore,

utilizing such haptic interactions for human-computer interaction is natural, and prior studies on force-based interaction methods [1–3, 5, 11] have shown its utility in enriching interactions. However, utilizing force requires a force sensor installed on the interaction surface, which limits its use to interacting with specialized input device. Several methods have been developed to enable indirect force sensing so that users can interact with any surfaces. For example, measuring the color changes of the finger nail using a camera [7, 12] or a nail-attached device [6, 8] could be used to estimate the pressing force. However, capturing the brightness and color changes of the finger nail using a camera is prone to ambient light changes, requires high-resolution camera for reliable force estimation, and needs per-user calibration [12]; using nail-attached devices require active sensors to be worn [6, 8].

We introduce Fringer (Figure 1), a passive finger-worn device that mechanically transforms force changes into visual displacement so that the force can be estimated by a normal webcam. Fringer uses Moiré patterns to magnify the displacement caused by the force interaction. Moiré patterns are produced when two periodic patterns are superimposed [9], which can amplify the feature along one dimension compared to the original pattern [4]. Research has shown that the Moiré patterns can be an ideal tool for measuring subtle changes, such as displacement [13–15] and deformation [10]. Inspired by these studies, Fringer detects pressing force by amplifying and visualizing the force change based on the displacements of fringe-like Moiré patterns created by two paper stripes. When a user wearing Fringer presses a surface, the compliant layer gets squeezed and pulls the printed paper stripe, which slides under the stationary stripe on its top and moves Moiré fringes. We also developed an image processing pipeline that detects the frame and Moiré fringes movements.

Fringer is small, non-intrusive, easy to put on and take off, and does not require active sensing components. It is also calibration-free and works with handheld digital cameras in various lighting conditions. In addition, the force detection range can easily be adjusted by varying compliant materials or grating periods.

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2 FORCE SENSING USING MOIRÉ FRINGES

On Fringer, we use two stripe patterns with different grating periods to create Moiré fringes that move along one direction (Figure 1). The grating period T_A of the top stripe is larger than the grating period T_B of the bottom (i.e. side) stripe. The touch force will be translated into the movement of the lower stripe while the top stripe remain stable. The grating period of the Moiré fringes T_M is calculated using Equation 1 [14]. T_M is always larger than T_A and T_B because of the division. If the stripe shift one full period, the Moiré fringes also move one whole period in the same direction, and therefore amplify the moving layer's displacement Δ_B .

$$T_M = \frac{T_A T_B}{T_A - T_B} \quad (1)$$

Δ_B is calculated from Moiré fringes' displacements Δ_M using Equation 2 [14]. The resolution of measurement is decided by T_M and the amplification ratio ($\frac{\Delta_M}{\Delta_B}$), which are controlled by adjusting T_A and T_B .

$$\Delta_B = \Delta_M \frac{T_A - T_B}{T_A} \quad (2)$$

The displacement caused by the compression of the silicone rubber is in the same direction as pressing is translated to Δ_B . Using Hooke's Law and the spring constant k of the silicone rubber, the force F produced by the fingertip is calculated using Equation 3.

$$F = k \Delta_B \quad (3)$$

3 IMPLEMENTATION OF FRINGER

3.1 Hardware Prototype

The main structure of Fringer is 3D printed including a top holder, a bottom sleeve, and a touch pad placed inside the sleeve (see Figure 1). A silicone rubber with a Shore hardness of 00-10 is placed under the touch pad to convert force into displacement. A white paper layer with laser-printed black strips is placed on a slide rail inside the holder and the fallen part is glued to the touch pad via a rounded edge. Therefore, the compression of the silicone rubber layer moves the paper with the stripe pattern.

The top layer with black strips is also made with paper, but the gaps between the strips are cut. This layer is immobile, glued to the holder, and placed on top of the printed layer. The grating period of the bottom and top stripes are 1.726 mm and 1.429 mm respectively, and they can produce Moiré fringes with a grating period of 8.305 mm. The center hollow part of the holder, which is framed by a white border, serves as a screen for Moiré fringes. A dark mark at the top left corner of the device helps the detection of the orientation of the screen. Fringer is a 32.8 mm × 19.8 mm × 34.8 mm finger-worn device, and weighs 8.6 g.

3.2 Image Processing Pipeline

The image processing pipeline first converts the live-fed image into grayscale and applies an adaptive threshold for contour detection using OpenCV. The detected contour as the inner rectangle of the white paper border is extracted. After locating the dark mark on the device, the contour's edges and corners are adjusted, and the screen area inside the selected contour is cropped. A following perspective transformation is conducted to rotate and rectify the screen.

Since we only need the displacement of the Moiré fringes along a single axis, we produce a single-pixel-height and 600-pixel-width grayscale image from the rectified and cropped screen area. A heavy Gaussian blur is applied to produce qualified fringes for calculation, similarly to the algorithm used in MoireBoard [14]. After the blur, the brightness distribution across the image will produce several inverted bell curves, the bottoms of which represent the centers of Moiré fringes. We continuously track the location and moving direction of the leftmost fringes and accumulate the displacements.

4 EVALUATION

To evaluate the relation between the pressing force and Moiré fringes, we put the Fringer on an electronic scale and used weights to apply different weights on the touch pad. An iPhone 13 Pro Max camera was used to capture the setup at 30 fps from a height of 40 cm. The image processing pipeline was running in real time to calculate the displacement of Moiré fringes. As Figure 2 shows, the linear regression result indicates that there exists a linear relation between the pressing force and detected Moiré fringe displacement ($r^2 = 0.9681$). On average, the image processing pipeline took 22.4 ms on a Macbook Pro with an M1 Pro processor.

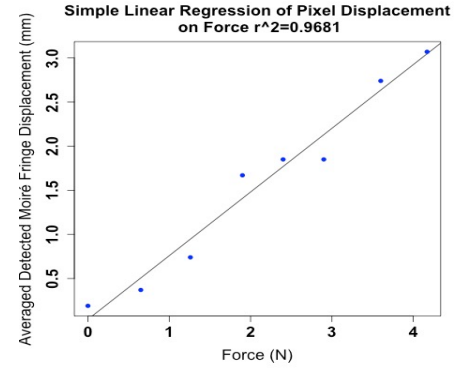


Figure 2: Simple Linear Regression of Detected Pixel Displacement on Pressing Force Applied

5 DISCUSSION AND FUTURE WORK

The evaluation shows that the pressing force could be successfully translated to the Moiré fringe movement within the force range of 0 - 4 N. This preliminary evaluation demonstrates Fringer's potential of measuring subtle pressing force change. To demonstrate potential applications of Fringer, we implemented two applications: a Flappy Bird game and a video capture program (see Figure 1(d,e)). These applications use the force measured by Fringer as a force-based video game controller and a haptic interaction force visualizer. Since the force can be detected as long as a camera captures the device, it can easily be used with any computing device with a camera, such as a laptop, mobile phone, and virtual reality headset.

The future work will include improving the image processing pipeline for more robust and accurate force detection, exploring different compliant materials, a thorough performance evaluation of the prototype, and a user study evaluating its usability.

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